Shashwat Solanki

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Education

PES University — Bengaluru, India
B.Tech in Computer Science & Engineering — *Aug 2023 - Present*
CGPA: 8.06

Experience

Research Intern - Space Applications Centre (ISRO) — Jul 2025 - Aug 2025
Geospatial Systems & Visualization — *Ahmedabad, India*

- Developed Vedant using CesiumJS, Vue.js, and ElectronJS to perform terrain profiling, viewshed analysis, and solar rooftop estimation on pan-India satellite datasets.
- Optimized 3D rendering pipeline and data handling for large raster datasets, improving performance by 30% during interactive geospatial visualization.
- Built RIDaM using Node.js and PostgreSQL for efficient storage, indexing, and retrieval of large-scale raster imagery and metadata.

Projects

Docksmith - Container Runtime System (Go)

- Built a Docker-like container system implementing image builds, layered storage, and process isolation.
- Designed and implemented a Dockerfile-like parser supporting FROM, COPY, RUN, WORKDIR, ENV, and CMD with validation.
- Implemented container runtime using Linux namespaces and chroot for process and filesystem isolation.
- Implemented content-addressed caching using SHA-256 hashes to reuse existing layers and avoid redundant builds.
- Built deterministic build system with reproducible tar normalization ensuring identical inputs generate identical outputs.

AI Reel Generator - Document to Video Pipeline (Next.js, Node.js, Flask, FFmpeg, Gemini API)

- Built an AI-powered pipeline converting PDFs and PPTs into short-form videos using LLM-based summarization.
- Designed backend workflow integrating document parsing, structured content extraction, and AI processing.
- Optimized processing pipeline for large documents improving overall rendering efficiency.

Vedant - Geospatial Analysis Tool (CesiumJS, Vue.js, ElectronJS)

- Integrated multiple geospatial analysis modules into a unified 3D simulation environment for large-scale satellite data.
- Optimized rendering pipeline and data handling, significantly improving performance for interactive visualization.

RIDaM - Raster Imagery Data Management (Node.js, PostgreSQL, Vue.js)

- Designed backend system for managing multi-source raster imagery datasets with efficient schema design and indexing.
- Implemented optimized query and storage strategies for handling large geospatial metadata.

Skills

Languages: C, C++, Python, JavaScript, Go
Backend: Node.js, Express.js, Flask
Frontend: React.js, Vue.js
Databases: PostgreSQL, MySQL, MongoDB
Systems: Linux, Process Isolation (chroot, namespaces), File Systems, Containerization Concepts
Tools: Git, GitHub, CI/CD

Certifications

- Introduction to Data Engineering on Google Cloud
- Geospatial Data Analysis Internship - ISRO
- Complete Python Bootcamp - Udemy
- C Programming Bootcamp - Udemy
- Problem Solving (Basic) - HackerRank

Achievements

- Secured 4th place in Labyrinth contest conducted by CodeChef at PES University among 60+ participants.
- Secured 6th place in Ignition hackathon conducted by Ather by developing a real-time sensor-based tracking device.